

Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8054

FACULTY OF SCIENCE AND TECHNOLOGY

F.E. (Group B) CSE / AI & ML / EEE (NEP) Sem - II

Examination November / December 2025 (To be held in February-2026)

Programming for Problem Solving (P-2024-25)

[Time: 2 Hours]

[Max. Marks: 30]

Please check whether you have got the right question paper.

N. B

1) Q.no. 1 is compulsory.

2) Solve any TWO questions from Q.no. 2, 3, 4 and 5.

SECTION A

Q1 Solve any five

10

A) What will be the output of the following code?

```
print(7 // 2)
```

- a) 3.5
- b) 4
- c) 3
- d) 2

B) What is the correct syntax for a basic if statement in Python?

- a) if x = 5:
- b) if (x == 5)
- c) if x == 5:
- d) if x = 5

C) What will be the output of the following code?

```
for i in range(1, 6):
```

```
if i == 3:
```

```
continue
```

```
print(i)
```

- a) 1 2
- b) 1 2 3
- c) 1 2 4 5
- d) 1 2 4

D) Which method adds a single element to the end of a list?

- a) add ()
- b) append ()
- c) insert ()
- d) extend ()

E) What is the output of the following code?

```
x = 10
```

```
if x > 5:
```

```
print("A")
```

```
elif x > 7:
```

```
print("B")
```

```
else:
```

`print("C")`

- a) C
 - b) B
 - c) A
 - d) A and B
- F) Can a tuple contain mutable elements?
- a) No
 - b) Yes
 - c) Only integers
 - d) Only strings
- G) Which method is used to delete a key from a dictionary?
- a) `pop ()`
 - b) `remove ()`
 - c) `discard ()`
 - d) `del ()`

SECTION B

- | | | |
|----|--|---|
| Q2 | A) Explain the features of Python programming language. | 5 |
| | B) What are Python identifiers? Write valid and invalid examples. | 5 |
| Q3 | A) With the help of syntax explain how to use <code>if ---else</code> statement in python also write program to read a number from user and check if it is positive or negative. | 5 |
| | B) Explain the use of <code>range ()</code> also write a program to display 1 to 10 even numbers. | 5 |
| Q4 | A) Explain recursive function. Write a code to display factorial of a number using recursive function. | 5 |
| | B) Write any five string built in methods with syntax and example program. | 5 |
| Q5 | A) What is slicing Write a Python program to show slicing a list using positive and negative indexing. | 5 |
| | B) Write a Python program to create and access elements in a tuple. | 5 |